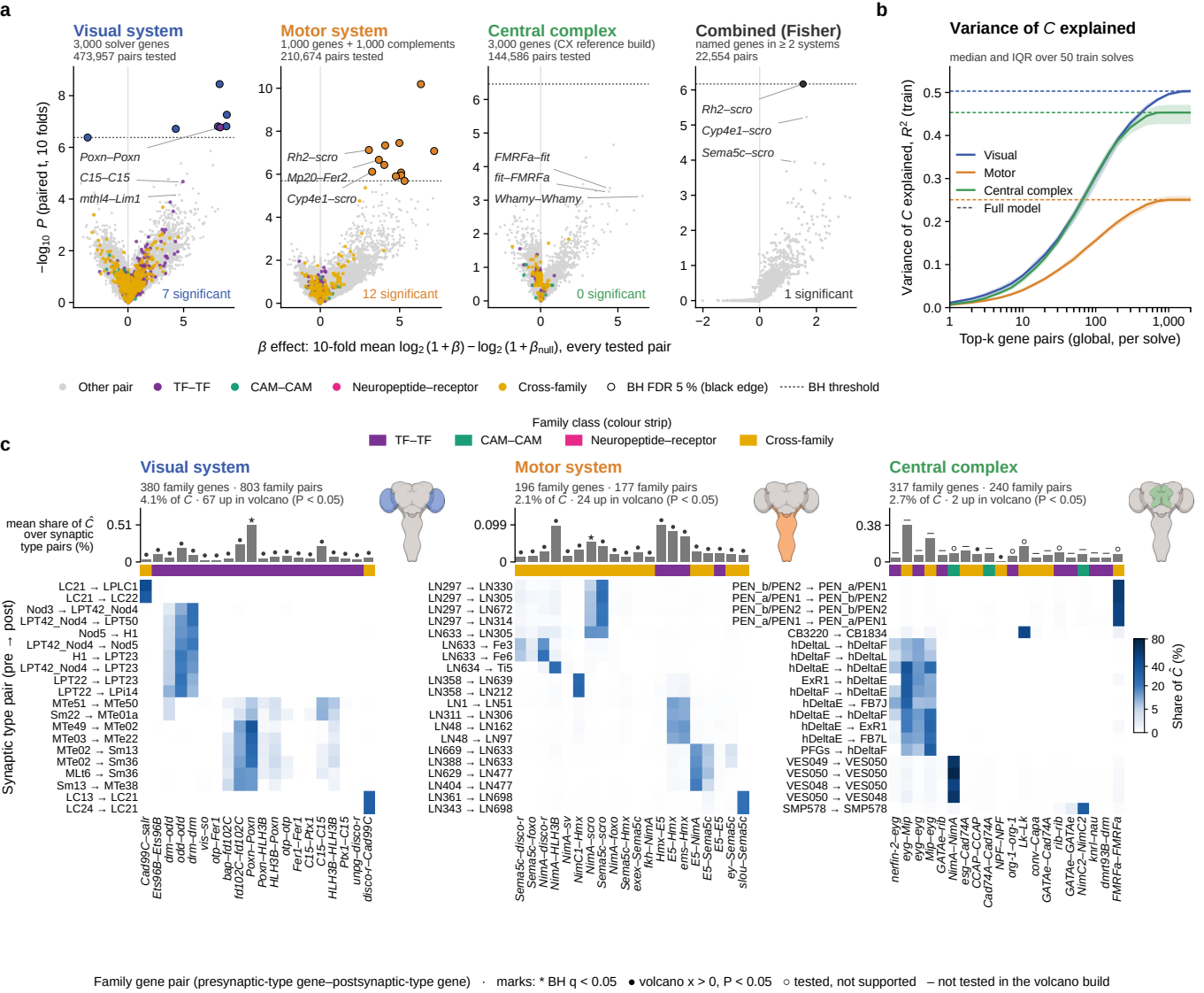

a
Visual system
3,000 solver genes
473,957 pairs tested
Motor system
1,000 genes + 1,000 complements
210,674 pairs tested
Central complex
3,000 genes (CX reference build)
144,586 pairs tested
Combined (Fisher)
named genes in ≥ 2 systems
22,554 pairs
−log10 P (paired t, 10 folds)
Poxn–Poxn
C15–C15
mthl4–Lim1
7 significant
Rh2–scro
Mp20–Fer2
Cyp4e1–scro
12 significant
FMRFa–fit
fit–FMRFa
Whamy–Whamy
0 significant
Rh2–scro
Cyp4e1–scro
Sema5c–scro
1 significant
β effect: 10-fold mean log2 (1 + β) − log2 (1 + βnull), every tested pair
Other pair
TF–TF
CAM–CAM
Neuropeptide–receptor
Cross-family
BH FDR 5 % (black edge)
BH threshold
b
Variance of C explained
median and IQR over 50 train solves
Variance of C explained, R2 (train)
Visual
Motor
Central complex
Full model
Top-k gene pairs (global, per solve)
c
Family class (colour strip)
TF–TF
CAM–CAM
Neuropeptide–receptor
Cross-family
Visual system
380 family genes · 803 family pairs
4.1% of Ĉ · 67 up in volcano (P < 0.05)
Motor system
196 family genes · 177 family pairs
2.1% of Ĉ · 24 up in volcano (P < 0.05)
Central complex
317 family genes · 240 family pairs
2.7% of Ĉ · 2 up in volcano (P < 0.05)
mean share of Ĉ over synaptic type pairs (%)
Synaptic type pair (pre → post)
Share of Ĉ (%)
Family gene pair (presynaptic-type gene–postsynaptic-type gene) · marks: * BH q < 0.05 ● volcano x > 0, P < 0.05 ○ tested, not supported – not tested in the volcano build